

# On covering radius of generalized Zetterberg codes

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# Notation

- $q_0$  is a prime power;
- $\mathbb{F}_{q_0}$  is the finite field with  $q_0$  elements;
- A subset  $\mathcal{C} \subseteq \mathbb{F}_{q_0}^n$  is called a code of length  $n$  over  $\mathbb{F}_{q_0}$ ;
- A vector  $\mathbf{x} \in \mathcal{C}$  is called a codeword;
- The Hamming distance of  $\mathbf{x} = (x_i)_{i=1}^n, \mathbf{y} = (y_i)_{i=1}^n \in \mathbb{F}_{q_0}^n$  is

$$d_H(\mathbf{x}, \mathbf{y}) = |\{i : x_i \neq y_i\}|;$$

- The minimum distance of  $\mathcal{C}$  is

$$d(\mathcal{C}) = \min_{\mathbf{c}_1 \neq \mathbf{c}_2 \in \mathcal{C}} d_H(\mathbf{c}_1, \mathbf{c}_2);$$

- The covering radius of  $\mathcal{C}$  is

$$\rho(\mathcal{C}) = \max_{\mathbf{x} \in \mathbb{F}_{q_0}^n} \min_{\mathbf{c} \in \mathcal{C}} d_H(\mathbf{x}, \mathbf{c}).$$

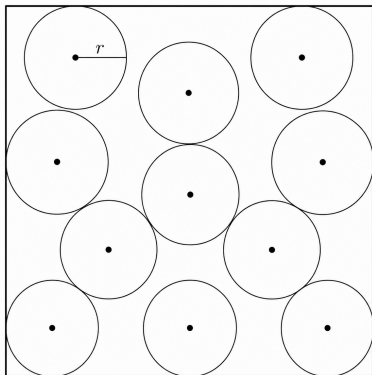
# Geometric interpretation

- $\mathbb{F}_{q_0}^n$  is a metric space with the Hamming metric  $d_H(\cdot, \cdot)$ .
- The Hamming sphere of radius  $r$  centered at  $\mathbf{x}$  is

$$B_{d_H}(\mathbf{x}, r) = \{\mathbf{y} \in \mathbb{F}_{q_0}^n : d_H(\mathbf{x}, \mathbf{y}) \leq r\}.$$

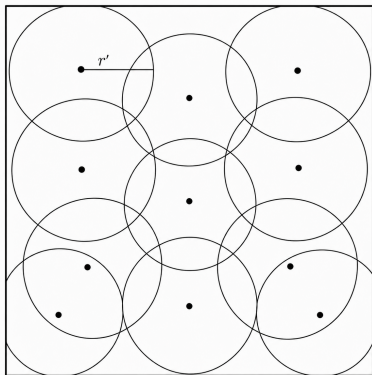
- The minimum distance  $d(\mathcal{C})$  and the covering radius  $\rho(\mathcal{C})$  are related to how to pack and cover the Hamming space  $\mathbb{F}_{q_0}^n$  by Hamming spheres of the same radius centered at codewords of  $\mathcal{C}$ .

# Minimum distance $d(\mathcal{C})$



**Figure:** The dots are the codewords of  $\mathcal{C}$ , and  $r = \lfloor (d-1)/2 \rfloor$  is the largest radius so that the Hamming spheres centered at  $\mathcal{C}$  do not overlap.

# Covering Radius $\rho(\mathcal{C})$



**Figure:** The dots are the codewords of  $\mathcal{C}$ ,  $r' = \rho(\mathcal{C})$  is the smallest radius such that the Hamming spheres centered at  $\mathcal{C}$  cover the whole space  $\mathbb{F}_{q_0}^n$ .

- The length  $n$ ,  $|\mathcal{C}|$ ,  $d(\mathcal{C})$  and  $\rho(\mathcal{C})$  are four fundamental parameters of  $\mathcal{C}$ .
- A basic bound:  $\rho(\mathcal{C}) \geq \left\lfloor \frac{d(\mathcal{C})-1}{2} \right\rfloor$ .
- A code is called *perfect* if  $\rho(\mathcal{C}) = \left\lfloor \frac{d(\mathcal{C})-1}{2} \right\rfloor$ ; it is called *quasi-perfect* if  $\rho(\mathcal{C}) = \left\lfloor \frac{d(\mathcal{C})-1}{2} \right\rfloor + 1$ .
- A code  $C$  is *maximal* if no supercode  $C' \supsetneq C$  exists with the same minimum distance. Equivalently, a maximal code satisfies

$$\rho(\mathcal{C}) \leq d(\mathcal{C}) - 1.$$

# Motivation and known results

- $\rho(C)$  is a fundamental parameter of  $\mathcal{C}$  and has many applications such as in decoding, data compression, etc.
- The covering radius problem has been studied extensively in the literature <sup>1</sup>
  - Bounds and exact determination of  $\rho(\mathcal{C})$ ;
  - Perfect codes and quasi-perfect codes;
  - On many interesting code families (BCH codes, Goppa codes, Reed-Muller codes, Reed-Solomon codes, Melas codes, etc);
  - Still many to come...
  - The most recent is related to generalized Zetterberg codes.

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<sup>1</sup>G. Cohen, I. Honkala, S. Litsyn, A. Lobstein, *Covering codes, ser.* North-Holland Mathematical Library. North-Holland Publishing Co., Amsterdam, 1997.



# Generalized Zetterberg codes

- $q_0$  is a prime power,
- $q = q_0^s$  where  $s \geq 2$  is a positive integer;
- 

$$H = U_{q+1} = \{x \in \mathbb{F}_{q^2} : x^{q+1} = 1\}.$$

- $H$  is a cyclic group of order  $q + 1$ . Let  $\xi$  be a generator. So

$$U_{q+1} = \{1, \xi, \xi^2, \dots, \xi^q\}$$

- The generalized Zetterberg code  $C_s(q_0)$  a cyclic code over  $\mathbb{F}_{q_0}$  of length  $q^s + 1$  and dimension  $q^2 + 1 - 2s$ , consisting of codewords of the form  $(c_0, c_1, \dots, c_q) \in \mathbb{F}_{q_0}^{q+1}$  such that

$$\sum_i c_i \xi^i = 0.$$

# Binary Zetterberg codes

- Binary Zetterberg code ( $q_0 = 2$ ) was introduced by Zetterberg<sup>2</sup> and Meggitt<sup>3</sup> independently in the 1960s.
- Code parameters  $[2^s + 1, 2^s + 1 - 2s]$  ( $s \geq 3$ ).

$$d = \begin{cases} 5 & \text{if } s \text{ is even;} \\ 3 & \text{if } s \text{ is odd,} \end{cases} \quad \rho = 3 \quad \text{if } s \geq 3.$$

- $s$  is even: quasi-perfect, reversible, etc
- There are efficient decoding algorithms.

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<sup>2</sup>L.-H. Zetterberg, "Cyclic codes from irreducible polynomials for correction of multiple errors," IRE Trans. Information Theory, vol. IT-8, pp. 13–20, 1962.

<sup>3</sup>J. Meggitt, "Error correcting codes and their implementation for data transmission systems," IRE Trans., vol. 7, no. 4, pp. 234–244, 1961. 

$C_s(q_0)$  has interesting relations to number theory

- the weight distribution of binary and ternary Zetterberg codes relates to Hecke operators acting on spaces of some modular forms,
- the number of weight 5 codewords connects to the Hasse zeta functions of certain K3 surfaces.
- these results played important role in recent work on the study of some Niho-type cross-correlations.

- A recent series of two papers provided a detailed account of  $\rho(C_s(q_0))$  for  $q_0$  odd<sup>4 5</sup>;
- the study is quite thorough, the argument is elementary but quite complex and intricate;
- the even  $q_0$  case is not treated;
- From number theory point of view, the covering radius problem of Zetterberg codes can be studied by standard techniques, for both  $q_0$  even and odd, in essentially the same way.

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<sup>4</sup>M. Shi, T. Hellesest and F. Özbudak, "Covering radius of generalized Zetterberg type codes over finite fields of odd characteristic", *IEEE Trans. Inform. Theory*, vol. 69, no. 11, pp. 7025-7048, 2023.

<sup>5</sup>M. Shi, S. Li, T. Hellesest and F. Öbudak, "Determining the covering radius of all generalized Zetterberg codes in odd characteristic", *IEEE Trans. Inform. Theory*, vol. 71, no. 5, pp. 3602-3613, 2025.

# Notation

- $q_0$  is a prime power,
- $q = q_0^s$  where  $s \geq 2$  is a positive integer;
- 

$$H = U_{q+1} = \{x \in \mathbb{F}_{q^2} : x^{q+1} = 1\}.$$

- $\bar{x} := x^q$  for  $x \in \mathbb{F}_{q^2}$ ;
- $C_s(q_0)$  is the generalized Zetterberg code of length  $q + 1$  over  $\mathbb{F}_{q_0}$ .

## Lemma (Lemma)

*The covering radius of  $C_s(q_0)$  is the least positive integer  $r$  such that for any  $\alpha \in \mathbb{F}_{q^2}$ , there exist  $c_1, \dots, c_\rho \in \mathbb{F}_{q_0}$  and  $x_1, \dots, x_\rho \in H$  such that*

$$\alpha = c_1 x_1 + \dots + c_\rho x_\rho.$$

Road-map of the argument:

- $r \neq 1$  (trivial);
- $r \leq 3$  (the major step);
- distinguish between  $r = 2$  and  $r = 3$  by careful analysis.
- we focus on that  $q_0$  is even.

$$\rho(C_s(q_0)) \leq 3$$

### Theorem (Theorem)

*For any  $\alpha \in \mathbb{F}_{q^2}$  and any fixed  $c_1, c_2, c_3 \in \mathbb{F}_{q_0}^*$ , the equation*

$$\alpha = c_1x_1 + c_2x_2 + c_3x_3$$

*is always solvable for  $x_1, x_2, x_3 \in H = U_{q+1}$ .*

Now we start the proof.

- May assume  $c_1 = c_2 = c_3 = 1$ .
- Let  $\alpha \in \mathbb{F}_{q^2}$ . Denote by  $N$  the number of  $x_1, x_2, x_3 \in H = U_{q+1}$  satisfying the equation

$$\alpha = x_1 + x_2 + x_3. \quad (1)$$

- the goal: to show that  $N \geq 1$ .
- Take  $q$ -th power on both sides of (1), noting  $x^q = 1/x$  if  $x \in H$ :

$$\alpha^q = x_1^{-1} + x_2^{-1} + x_3^{-1}. \quad (2)$$

- we eliminate variable  $x_3$  from (1) and (2) to obtain



- we eliminate variable  $x_3$  from (1) and (2) to obtain

$$(\alpha^q - x_1^{-1})x_2^2 - ((\alpha - x_1)(\alpha^q - x_1^{-1}))x_2 + (\alpha - x_1) = 0. \quad (3)$$

Eq (3) is a quadratic equation which is well understood:

For any fixed  $x_1 \in H$  with  $\alpha^q - x_1^{-1} \neq 0$ , (3) is solvable (with exactly two solutions) for  $x_2 \in H$  if and only if

$$q_0 \text{ even : } \operatorname{Tr}_{\mathbb{F}_q/\mathbb{F}_2} \left( \frac{(\alpha^q + \bar{x}_1)(\alpha + x_1)}{((\alpha + x_1)(\alpha^q + \bar{x}_1))^2} \right) = 1.$$

$$q_0 \text{ odd : } ((\alpha - x_1)(\alpha^q - x_1^{-1}))^2 - 4(\alpha^q - x_1^{-1})(\alpha - x_1) \\ \text{is a non-square in } \mathbb{F}_q.$$

Denote

$$f(x_1) = \frac{(\alpha^q + \bar{x}_1)(\alpha + x_1)}{((\alpha + x_1)(\alpha^q + \bar{x}_1))^2}, \quad q_0 \text{ even};$$

$$\begin{aligned} \Delta(x_1) = & ((\alpha - x_1)(\alpha^q - x_1^{-1}))^2 \\ & - 4(\alpha^q - x_1^{-1})(\alpha - x_1), \quad q_0 \text{ odd}. \end{aligned}$$

Then for any given  $x_1 \in H \setminus \{\alpha\}$ , the number of  $x_2 \in H$  satisfying (3) is

$$\begin{aligned} 1 - \Phi(f(x_1)) : & \quad q_0 \text{ even} \\ 1 - \chi(\Delta(x_1)) : & \quad q_0 \text{ odd} \end{aligned}$$

where

$q_0$  even :  $\Phi(x) = (-1)^{\text{Tr}_{\mathbb{F}_q/\mathbb{F}_2}(x)}$  is the additive character on  $\mathbb{F}_q$ ;

$q_0$  odd :  $\chi$  is the quadratic character on  $\mathbb{F}_q^*$ .

So we have

- if  $q_0$  is even, then

$$N \geq \sum_{x_1 \in H \setminus \{\alpha\}} 1 - \Phi(f(x_1)) = q - \sum_{x_1 \in H \setminus \{\alpha\}} \Phi(f(x_1));$$

- if  $q_0$  is odd, then

$$N \geq \sum_{x_1 \in H \setminus \{\alpha\}} 1 - \chi(\Delta(x_1)) = q - \sum_{x_1 \in H \setminus \{\alpha\}} \chi(\Delta(x_1)).$$

# Character sum over $H$

By using the bijection  $\eta : \mathbb{P}^1(\mathbb{F}_q) \rightarrow H = U_{q+1}$  given by

$$\eta : x \mapsto \frac{x + \xi}{x + \xi q}, \quad \infty \mapsto 1,$$

we can write

- if  $q_0$  is even, then

$$\sum_{x_1 \in H \setminus \{\alpha\}} \Phi(f(x_1)) \sim \sum_{x \in \mathbb{F}_q} \Phi(f(\eta(x))) = \sum_{x \in \mathbb{F}_q} \Phi(f'(x))$$

- if  $q_0$  is odd, then

$$\sum_{x_1 \in H \setminus \{\alpha\}} \chi(\Delta(x_1)) \sim \sum_{x \in \mathbb{F}_q} \chi(\Delta(\eta(x))) = \sum_{x \in \mathbb{F}_q} \chi(\Delta'(x)),$$

where  $f', \Delta' \in \mathbb{F}_q(X)$  are explicitly given rational functions.

# The Weil-type bound

By applying the standard Weil-type bound

$$\left| \sum_{x \in \mathbb{F}_q} \Phi(f'(x)) \right| \leq C\sqrt{q}, \quad \left| \sum_{x \in \mathbb{F}_q} \chi(\Delta'(x)) \right| \leq C\sqrt{q},$$

for both  $q_0$  even and odd, where the constant  $C$  is explicitly given, we have

$$N \geq q - C\sqrt{q} > 0 \quad \text{if } q > C^2.$$

When  $q = q_0^s \leq C^2$ , the conclusion  $\rho(C_s(q_0)) \leq 3$  can be verified by using Magma.

# Determining $\rho(C_s(q_0)) \in \{2, 3\}$

## Lemma

- $\rho(C_s(q_0)) = 3$  if and only if there exists  $\alpha \in \mathbb{F}_{q^2}$  such that the equation

$$\alpha = c_1x_1 + c_2x_2 \tag{4}$$

is not solvable for any  $c_1, c_2 \in \mathbb{F}_{q_0}$  and any  $x_1, x_2 \in H$ .

- $\rho(C_s(q_0)) = 2$  if and only if for any  $\alpha \in \mathbb{F}_{q^2}$ , Eq (4) is solvable for  $c_i \in \mathbb{F}_{q_0}, x_i \in H$ .

# Ideas

Just careful analysis:

- Eliminate one variable say  $x_2$  to get a quadratic equation, from which we obtain a precise criterion as to whether it is solvable or not for  $x_1 \in H$ ;
- We need to keep track of all  $c_1, c_2 \in \mathbb{F}_{q_0}$ , the conditions can be written explicitly;
- Just do a careful analysis, try to get rid of  $c_1, c_2 \in \mathbb{F}_{q_0}$ .

# Criterion

## Theorem

*Let  $q_0 \geq 2$  be even and  $s \geq 2$ . Then  $\rho(C_s(q_0)) = 3$  if and only if there exists  $\alpha \in \mathbb{F}_q \setminus \mathbb{F}_{q_0}$  such that*

*•  $\text{Tr}_{\mathbb{F}_q/\mathbb{F}_{q_0}}(\alpha) = 0$ , and*

*•  $\text{Tr}_{\mathbb{F}_q/\mathbb{F}_{q_0}}\left(\frac{1}{1+b\alpha}\right) \in \{0, 1\}$  for all  $b \in \mathbb{F}_{q_0}^*$ .*

*Otherwise  $\rho(C_s(q_0)) = 2$ .*



# Main result $q_0$ is even

## Theorem

*Let  $q_0$  be even. The covering radius of  $C_s(q_0)$  is at most three and is given by*

- i). *If  $s = 1$ , then  $\rho(C_s(q_0)) = 1$ ;*
- ii). *If  $s = 2$ , then  $\rho(C_s(q_0)) = 2$ ;*
- iii). *If  $s \geq 4$  is even, then  $\rho(C_s(q_0)) = 3$ ;*
- iv). *If  $s \geq 3$  is odd, and  $s \leq s^*(q_0)$ , then  $\rho(C_s(q_0)) = 2$ ;*
- v). *If  $s \geq 3$  is odd, and  $s \geq s_*(q_0)$ , then  $\rho(C_s(q_0)) = 3$ ;*
- vi). *If  $s \geq 3$  is odd, and  $s' | s$  with  $\rho(C_{s'}(q_0)) = 3$ , then  $\rho(C_s(q_0)) = 3$ .*

# Main result $q_0$ is even

## Theorem

Here  $s_*(q_0) \geq 3$  (if exists) is the least odd number  $s$  satisfying the inequality

$$q_0^s - q_0^{s/2} \left(\frac{q_0}{2}\right)^{q_0-1} (2q_0^2 - 7q_0 + 3) > \left(\frac{q_0}{2}\right)^{q_0-1} (4q_0 - 2) - q_0^2,$$

and  $s^*(q_0) \geq 3$  (if exists) is the largest odd number satisfying  $s^*(q_0) \leq q_0/2$ .

## Table

Table:  $s^*(q_0), s_*(q_0)$  and values not covered.

| $q_0 = 2^t$ | $s^*$ | $s_*$ | the set of missing odd numbers in $(s^*, s_*)$ |
|-------------|-------|-------|------------------------------------------------|
| 2           |       | 3     | $\emptyset$                                    |
| 4           |       | 7     | $\{3, 5\}$                                     |
| 8           | 3     | 15    | $\{5, 7, 9, 11, 13\}$                          |
| 16          | 7     | 27    | odd numbers from 9 to 25                       |
| 32          | 15    | 55    | odd numbers from 17 to 53                      |
| 64          | 31    | 111   | odd numbers from 33 to 109                     |
| 128         | 63    | 223   | odd numbers from 65 to 221                     |

# The End

Thank you!